

Class #4 - Eigenvectors and Linear Transformations

1. Example: Let $B = \{\vec{b}_1, \vec{b}_2, \vec{b}_3\}$ and $D = \{\vec{d}_1, \vec{d}_2\}$ be bases for vector spaces V and W , respectively. Let $T: V \rightarrow W$ be a linear transformation with the property that T maps $\vec{b}_1$ to $3\vec{d}_1 - 5\vec{d}_2$, $\vec{b}_2$ to $-\vec{d}_1 + 6\vec{d}_2$, and $\vec{b}_3$ to $4\vec{d}_2$.

$$\begin{cases} T(x+y) = T(x) + T(y) \\ T(cx) = cT(x) \end{cases}$$

$$[T(\vec{b}_i)]_D = \begin{bmatrix} 3 \\ -5 \\ 0 \end{bmatrix}, \text{ etc.}$$

$$T(\vec{b}_1) = 3\vec{d}_1 - 5\vec{d}_2$$

$$T(\vec{b}_2) = -\vec{d}_1 + 6\vec{d}_2$$

$$T(\vec{b}_3) = 4\vec{d}_2$$

$$\dim V = 3$$

$$\dim W = 2$$

$$T: V \rightarrow W$$

Take any $x \in V$

$$\text{Then, } x = c_1\vec{b}_1 + c_2\vec{b}_2 + c_3\vec{b}_3 \text{ for some scalars } c_1, c_2, c_3.$$

$$\begin{aligned} T(x) &= T(c_1\vec{b}_1 + c_2\vec{b}_2 + c_3\vec{b}_3) \\ &\stackrel{(T \text{ linear})}{=} c_1 T(\vec{b}_1) + c_2 T(\vec{b}_2) + c_3 T(\vec{b}_3) \\ &= c_1 (3\vec{d}_1 - 5\vec{d}_2) + c_2 (-\vec{d}_1 + 6\vec{d}_2) + c_3 (4\vec{d}_2) \end{aligned}$$

$$= (3c_1 - c_2)\vec{d}_1 + (-5c_1 + 6c_2 + 4c_3)\vec{d}_2$$

$$\Rightarrow [T(x)]_D = \begin{bmatrix} 3c_1 - c_2 \\ -5c_1 + 6c_2 + 4c_3 \end{bmatrix} = \begin{bmatrix} 3 & -1 & 0 \\ -5 & 6 & 4 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

$$\therefore [T(x)]_D = \begin{bmatrix} [T(\vec{b}_1)]_D & [T(\vec{b}_2)]_D & [T(\vec{b}_3)]_D \end{bmatrix} [x]_B$$



$$\dim V = n$$

$$\dim W = m$$

2. In general, let V and W be two finite dimensional vector spaces with dimensions n and m , respectively. Let $T: V \rightarrow W$ be any linear transformation. Let B and C be bases for V and W , respectively. Given any $x \in V$, the coordinate vector $[x]_B$ is in $\mathbb{R}^n$. Then, the image $T(x)$ is in W and its coordinate vector $[T(x)]_C$ is in $\mathbb{R}^m$.

$$\text{Hence, } [T(x)]_C = \begin{bmatrix} [T(\vec{b}_1)]_C & \dots & [T(\vec{b}_n)]_C \end{bmatrix} [x]_B.$$

The matrix is a matrix representation of T , and is called the matrix for T relative to the bases B and C .

① Attendance

② Participation Quiz

③ Class #4 handout ~

④ Will discuss Q #1 flight: 10
Aug. 9.26

⑤ Q #2 Mon. PP #4 - #5

⑥ Office hr today 3-5 pm

$\Rightarrow$ 3 - 3:50 pm & 1:20 - 2:30 pm
(to day) (Mon)

due to a seminar

(T/F)

① If A is similar to B , then B is similar to A . (T)
 $A = P^{-1} B P$
 $P^{-1} A P = B$

② Suppose A is 5×5 matrix.

$\lambda_1, \lambda_2, \lambda_3$ distinct eigenvalues.

a) (T/F) A has at least 4 L.I. eigenvectors. (F)

b) (T/F) possible geo. mult. of λ_3 is
~~0, 1, 2, 3.~~ (F)

c) (T/F) possible geo. mult. of λ_3 is
1, 2, 3, 4. (F)

d) (T/F) possible alg. mult. of λ_3 is 1, 2, 3. (T)

$$A \in \mathbb{R}^{5 \times 5}$$

(3) A has 5 distinct eigenvalues.

(a) (T/P) A is similar to a diagonal matrix. (T)

(b) (T/P) A is diagonalizable. (T) $AP = PD$

(c) (T/P) There is a basis of $\mathbb{R}^5$ that consists of eigenvectors of A only. (T) $\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_5 \end{bmatrix} \begin{bmatrix} v_1 & & \\ & v_2 & \\ & & v_5 \end{bmatrix}$

(4) $\mathbb{R}^2$. $B = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$

$$x = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + (-3) \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Coordinate vector of x relative to Basis B

$$[x]_B = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$$

$$\{ a_0 + a_1 t + \dots + a_n t^n \}$$

(5) P_3 : the set of all polynomials of degree ≤ 3 .

the standard basis of $P_3 = \{ 1, t, t^2, t^3 \}$

(6) V : vector space
one Basis = $\{ \vec{b}_1, \dots, \vec{b}_{10} \}$

(T/P) $\dim V = 9, 10$

3. Example: Define $T : \mathbf{P}_3 \rightarrow \mathbf{R}^4$ by $T(p) = \begin{bmatrix} p(-3) \\ p(-1) \\ p(1) \\ p(3) \end{bmatrix}$.

$$\textcircled{ex} \quad T(1 + 3t) = \begin{bmatrix} 1+3(-3) \\ 1+3(-1) \\ 1+3(1) \\ 1+3(3) \end{bmatrix}$$

(a) Show that T is a linear transformation.

$$T(p_1(t) + p_2(t)) = T\left(\underbrace{(a_0 + a_1 t + a_2 t^2 + a_3 t^3)}_{\text{polynomial}} + \underbrace{(b_0 + b_1 t + b_2 t^2 + b_3 t^3)}_{\text{polynomial}}\right)$$

$$= T(p_1(t) + p_2(t))$$

$$= \begin{bmatrix} p_1(-3) + p_2(-3) \\ p_1(-1) + p_2(-1) \\ p_1(1) + p_2(1) \\ p_1(3) + p_2(3) \end{bmatrix} = \begin{bmatrix} p_1(-3) \\ p_1(-1) \\ p_1(1) \\ p_1(3) \end{bmatrix} + \begin{bmatrix} p_2(-3) \\ p_2(-1) \\ p_2(1) \\ p_2(3) \end{bmatrix}$$

$$= T(p_1(t)) + T(p_2(t))$$

$$T(c p(t)) = \begin{bmatrix} c p(-3) \\ c p(-1) \\ c p(1) \\ c p(3) \end{bmatrix} = c \begin{bmatrix} p(-3) \\ p(-1) \\ p(1) \\ p(3) \end{bmatrix} = c T(p(t))$$

$$[T(\vec{b}_i)]_C = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$T(\vec{b}_1) = T(p(t)=1) = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = 1 \cdot \vec{c}_1 + 1 \cdot \vec{c}_2 + 1 \cdot \vec{c}_3 + 1 \cdot \vec{c}_4$$

(b) Find the matrix for T relative to the standard basis for $\mathbf{P}_3$ and the standard basis for $\mathbf{R}^4$.

$$B = \{1, t, t^2, t^3\}$$

$$C = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$[T(x)]_C = \begin{bmatrix} [T(\vec{b}_1)]_C & \dots & [T(\vec{b}_4)]_C \end{bmatrix} [x]_B$$

Matrix for T relative to the Bases B and C

$$= \begin{bmatrix} 1 & -3 & 9 & -27 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & 1 & 1 \\ 1 & 3 & 9 & 27 \end{bmatrix}$$

① Q #2

② 5 min. break

③ Attendance

④ Par. quiz

⑤ Class #4 #4 ~

⑥ Quiz #3 Mon 9/28 (PP #6 ~ #7)

⑦ Video Lecture for 9/25 (holiday)

(T/F)

① Sometimes, a real matrix can have complex eigenvalues.



$$\begin{bmatrix} * & * \\ * & * \end{bmatrix}$$

Matrix for T relative to the bases

② (T/F) If a real matrix has a complex eigenvalue $1+i$ and $1-i$ is also an eigenvalue. (T)

③ Name of

$$[A]x = \lambda x$$

$$\textcircled{1} \dim W = \boxed{4}$$

③ (T/F) If a real matrix has $1+i$ as an eigenvalue, then the corresponding eigenvector is also complex. (T)

④ Basis of V : A

⑤ Basis of W : D

$$\textcircled{2} \dim V = \boxed{5}$$

④

T (linear transformation) : $V \rightarrow W$
in $\mathbb{R}^4$ V and W are vector spaces

$$[T(x)] = \begin{bmatrix} [T(e_1)] & \dots & [T(e_5)] \end{bmatrix} [x]_A$$

4. Now suppose we have a finite dimensional vector space V with a basis B . If $T: V \rightarrow V$ is a linear transformation, then the matrix representation of T is called the matrix for T relative to B , or simply the B -matrix for T , and is denoted as $[T]_B$.

$$[T(x)]_B = \begin{bmatrix} [T(\vec{b}_1)]_B & \dots & [T(\vec{b}_n)]_B \end{bmatrix} [x]_B$$

$n \times n$

5. Example: Let $B = \{\vec{b}_1, \vec{b}_2, \vec{b}_3\}$ be a basis for a vector space V . Find $T(3\vec{b}_1 - 4\vec{b}_2)$ when T is a linear transformation from V to V whose matrix relative to B is

B -matrix for T

$$[T]_B = \begin{bmatrix} 0 & -6 & 1 \\ 0 & 5 & -1 \\ 1 & -2 & 7 \end{bmatrix}$$

$$[T(\vec{b}_1)]_B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$T(\vec{b}_1) = 0 \cdot \vec{b}_1 + 0 \cdot \vec{b}_2 + 1 \cdot \vec{b}_3$$

$$T(\vec{b}_1) = \vec{b}_3$$

$$B = \{\vec{b}_1, \vec{b}_2, \vec{b}_3\}$$

$$[x]_B = \begin{bmatrix} 5 \\ 7 \\ 7 \end{bmatrix}$$

$$\Leftrightarrow x = 5 \cdot \vec{b}_1 + 0 \cdot \vec{b}_2 + 7 \cdot \vec{b}_3$$

$$T(3\vec{b}_1 - 4\vec{b}_2)$$

$$= 3T(\vec{b}_1) - 4T(\vec{b}_2)$$

(linear)

$$= 3 \cdot (\vec{b}_3) - 4(-6\vec{b}_1 + 5\vec{b}_2 - 2\vec{b}_3)$$

$$= 24\vec{b}_1 - 20\vec{b}_2 + 11\vec{b}_3$$

6. Note that the matrix representation of T depends on the choice of basis vectors. Can we find a basis of V so that $[T]_B$ becomes a diagonal matrix?

Example: Define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(x) = Ax$. Find a basis B for $\mathbb{R}^2$ with the property that $[T]_B$ is diagonal.

(a)

? B

$$A = \begin{bmatrix} 4 & -2 \\ -1 & 3 \end{bmatrix}$$

$$\det(A - \lambda I)$$

$$= \det \begin{bmatrix} 4-\lambda & -2 \\ -1 & 3-\lambda \end{bmatrix}$$

$$= (4-\lambda)(3-\lambda) - 2$$

$$= \lambda^2 - 7\lambda + 10$$

$$= (\lambda-2)(\lambda-5)$$

$$\text{Nul}(A - 2I) = \text{Nul} \left(\begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix} \right) = \text{Nul} \left(\begin{bmatrix} -1 & -2 \\ -1 & -2 \end{bmatrix} \right)$$

$\vec{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$$[T]_B = \begin{bmatrix} [A\vec{b}_1]_B & [A\vec{b}_2]_B \\ [T(\vec{b}_1)]_B & [T(\vec{b}_2)]_B \end{bmatrix} = \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix}$$

Find $B = \{\vec{b}_1, \vec{b}_2\}$

$\therefore$ Want $\vec{b}_1$ and $\vec{b}_2$ such that

$$[A\vec{b}_1]_B = \begin{bmatrix} \alpha \\ 0 \end{bmatrix} \quad \text{and} \quad [A\vec{b}_2]_B = \begin{bmatrix} 0 \\ \beta \end{bmatrix}$$

$$\Leftrightarrow A\vec{b}_1 = \alpha\vec{b}_1 + 0\vec{b}_2 \quad \text{and} \quad A\vec{b}_2 = 0\vec{b}_1 + \beta\vec{b}_2$$

$$\Leftrightarrow A\vec{b}_1 = \alpha\vec{b}_1 \quad \text{and} \quad A\vec{b}_2 = \beta\vec{b}_2$$

α, β scalars
 $\vec{b}_i \neq 0 \quad i=1,2$

$\therefore B$ consists of L.I. eigenvectors.

$$\therefore [T]_B = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$$

$$\text{with eigenvector basis } \left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\}$$

...

(b)

$$A = \begin{bmatrix} 1 & 1 \\ -1 & 3 \end{bmatrix}$$

$$\det(A - \lambda I) = (\lambda - 2)^2 = 0$$

$$\lambda = 2, 2 \quad \text{1 L.I. column.} \quad \therefore \text{rank} = 1.$$

$\text{Nul}(A - 2I) = \text{Nul}\left(\begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}\right)$
 $B = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$
 Can fit in with any L.I. vector as this eigenvector.
 (But $\perp$ is the best.)
 $[T]_B$ is not diagonal. Geomult. = nullity $(A - 2I) = 1$
 There is no Basis such that $[T]_B$ is diagonal!
 eigenvalue: $\lambda = 2$
 eigenspace: $\mathbb{R}^2$